

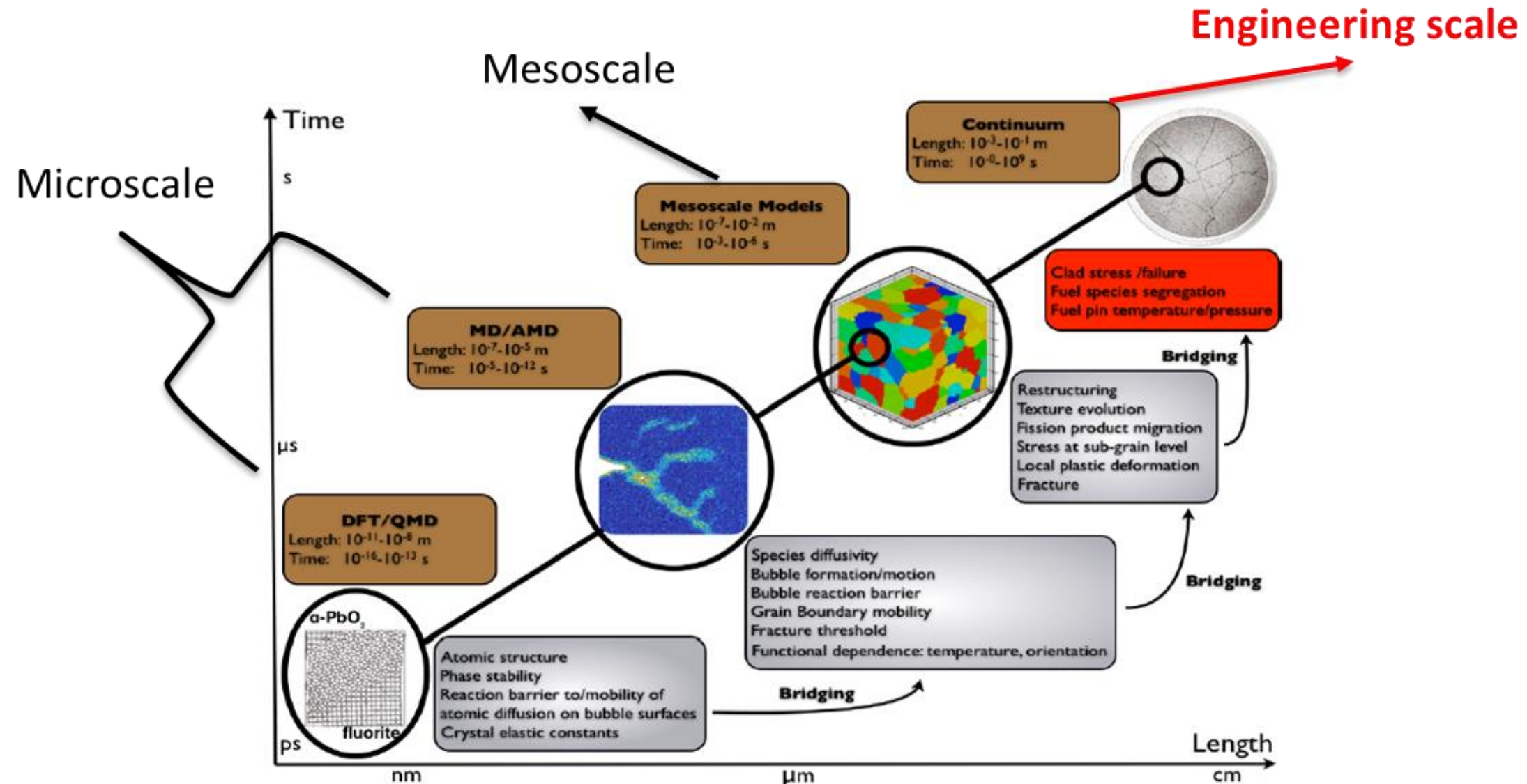
NE 511: Multi-Physics of Nuclear Reactors

UNIT 3: Low Fidelity Multi-Physics Modeling

***Lecture 15: FUEL PERFORMANCE WITH NEUTRONICS AND THERMAL-HYDRAULICS FEEDBACK
HEAT CONDUCTION AND FUEL ROD MODELING IN SUBCHANNEL AND SYSTEM TH CODES***

OVERVIEW

Fuel modeling covers various degrees of scales, from atomistic to full core, and from picoseconds to years

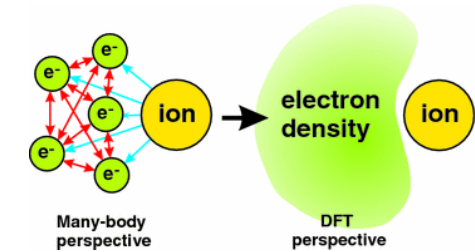


OVERVIEW

Microscale modeling: the realm of Physicists

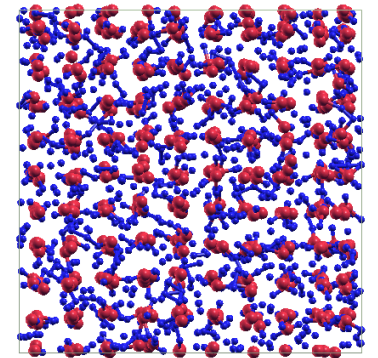
Density Functional Theory (DFT):

- Based on Quantum Theory without any empirical parameter
- First-principles electronic structure calculations
- Schrödinger equation of interactive electrons is transformed to a single-electron equivalent expression that moves in an effective potential
- The total energy is minimized and quantities such as the neutron density and the effective potential are calculated



Classical Molecular Dynamics (CMD):

- Evolution of molecules under force fields
- Empirical potentials are used to compute the force fields
- Well-suited for computing irradiation damage, fission gas production and release and evolution of materials properties
- Limitation due to the way the empirical potentials are obtained (experimental data or DFT)
- Can be used to provide a bridge between DFT and Mesoscale models or Engineering scale models



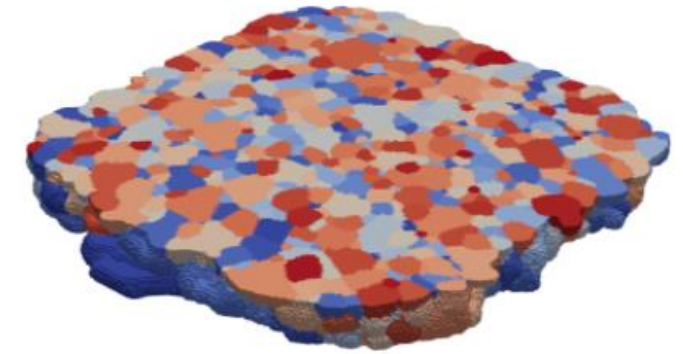
- **DFT tries to find the total energy of the system using electron density. This can be used to compute the electronic structure, potentials and associated properties.**
- **CMD on the other side track the dynamic movement of molecules and atoms in a force field using empirical potentials.**

OVERVIEW

Mesoscale modeling: the realm of Materials Scientists

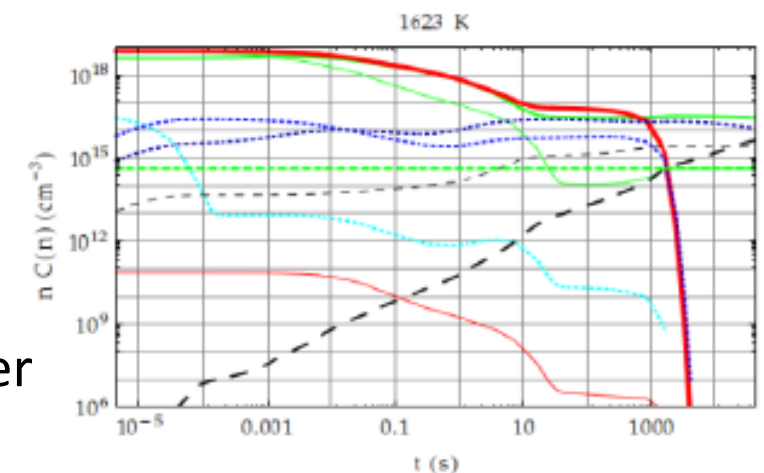
Phase Field:

- Individual phases and their interphases are described by a set of parameters
- Microstructure evolution is computed by calculating the time-dependent parameters and concentrations that minimize the system energy
- Irradiation effects such as grain growth and hydrides precipitation in cladding have been modeled



Cluster Dynamics:

- Based on rate theory
- Concentration evolution of clusters such as point defects and soluble atoms
- Resembles the Bateman equations
- Computationally very cheap but spatial details loss within every cluster



OVERVIEW

Fuel rod scale modeling: the realm of Nuclear Engineers

High Fidelity:

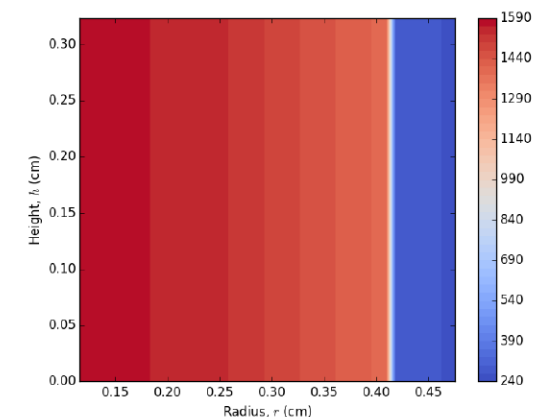
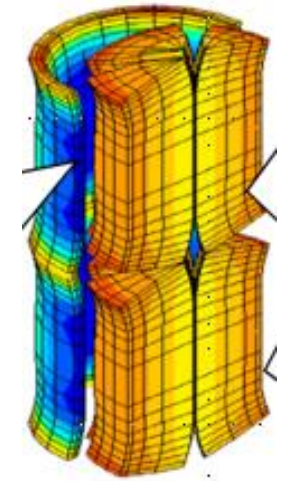
- Finite elements 3D/2D or 1.5D
- Full coupling of thermal, mechanical and physico-chemical aspects
- Empirical models for most important phenomena
- Computationally expensive and mainly used in R&D

Standard Fidelity:

- Finite differences 1.5D
- Full coupling of thermal, mechanical and physico-chemical aspects
- Empirical models for most important phenomena
- Computationally very efficient and used in both R&D and licensing

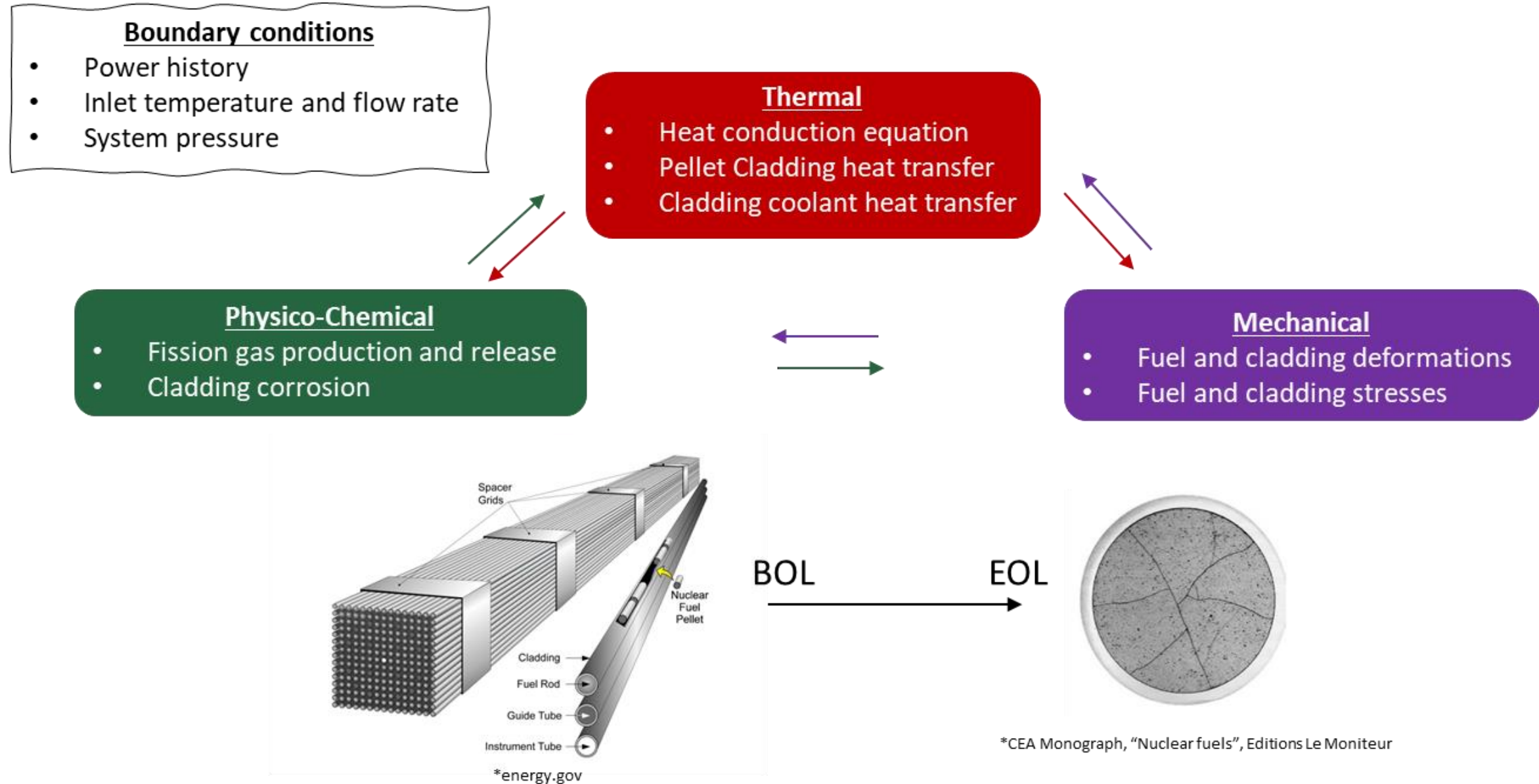
Low Fidelity:

- Finite differences 1D
- Solve only heat conduction in fuel and cladding by imposing the gap heat transfer
Few thermal phenomena are considered
- Computationally very cheap and usually consist thermal solvers of thermal-hydraulics codes



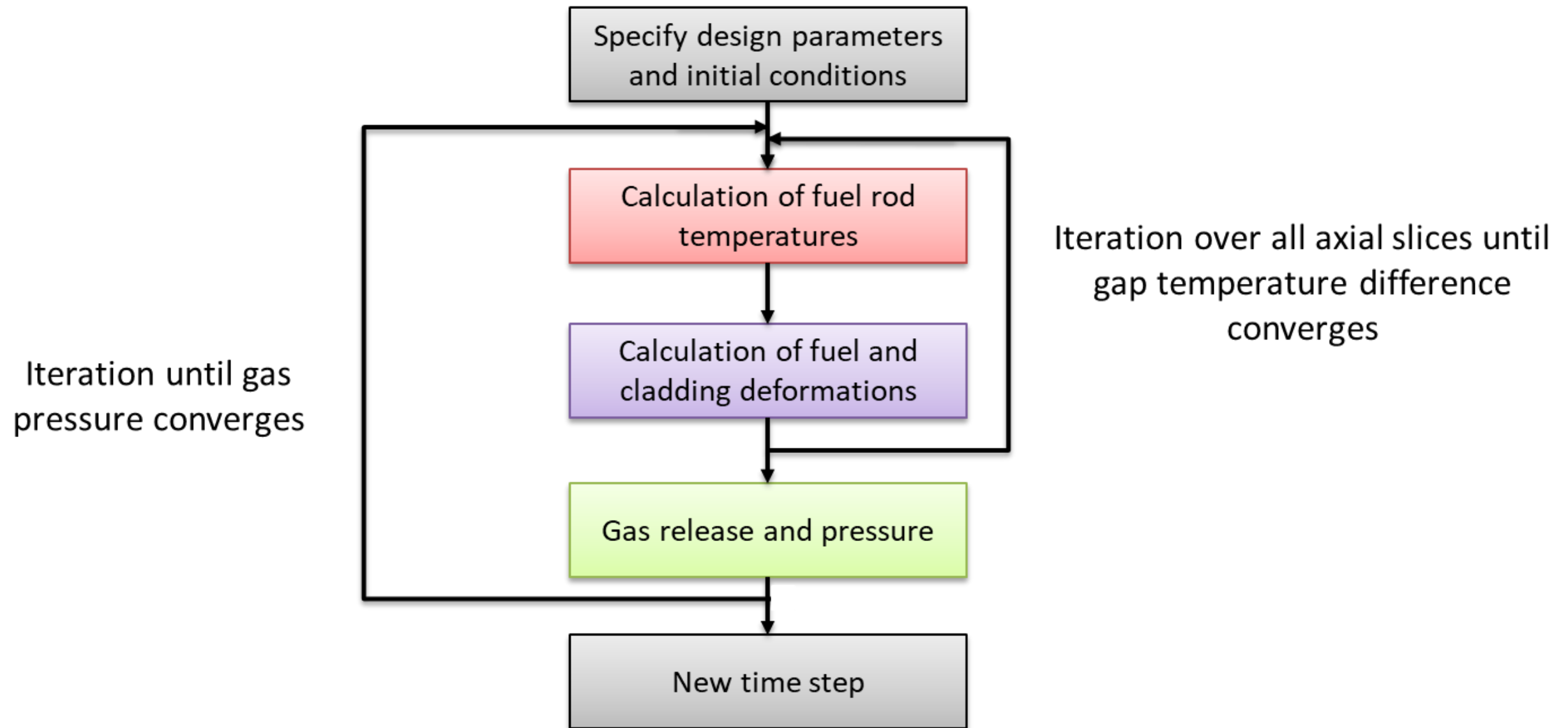
OVERVIEW

Irradiation Phenomena from BOL to EOL



OVERVIEW

1.5D coupling between thermal, mechanical and physico-chemical modelling



FUEL PERFORMANCE WITH NEUTRONICS AND THERMAL-HYDRAULICS FEEDBACK

NEUTRONICS MODELS IN FUEL PERFORMANCE CODES

Example: BISON

- “ ... BISON is a finite element-based nuclear fuel performance code applicable to a variety of fuel forms including LWR fuel rods, TRISO particle fuel, and metallic rod and plate fuel
- It solves the fully-coupled equations of thermo-mechanics and species diffusion, for either 1D spherical, 2D axisymmetric or 3D geometries
- Fuel models: temperature and burnup dependent thermal properties, fission product swelling, densification, thermal and irradiation creep, fracture, and fission gas production and release
- Clad models: plasticity, irradiation growth, and thermal and irradiation creep models
- Additional models: gap heat transfer, mechanical contact, and the evolution of the gap/plenum pressure with plenum volume, gas temperature, and fission gas addition
- BISON is based on the MOOSE framework and can therefore efficiently solve problems using standard workstations or very large high-performance computers. ...”

NEUTRONICS MODELS IN FUEL PERFORMANCE CODES

Power Model in BISON (TUBRNP model of Lassmann)

!! TUBRNP Models is also used in FRAPCON/FRAPTRAN & TRANSURANUS !!

Total Power

- *The power associated with an LWR fuel rod is typically given as rod averaged linear power (or linear heat rate) in units of W/m*
- *Power can vary in time and space*

Axial Power Profile

- *Given as a scaling factor, which is a function of distance from the bottom of the rod*

Radial Power Profile

- *Function of geometry, initial fuel characteristics, and irradiation history:*

Increasing burnup → high Pu concentration toward the outer rim → 2-3 times higher local power density

NEUTRONICS MODELS IN FUEL PERFORMANCE CODES

TUBRNP model

$$q'''(r) = \alpha \sum_k \sigma_{f,k} N_k \phi$$

Average isotope concentration is:

$$dbu = \frac{q''' dt}{\rho_{fuel}} = \frac{\alpha}{\rho_{fuel}} \sum_k \sigma_{f,k} N_k \phi dt$$

$$\frac{d\bar{N}_{235}}{dt} = -\sigma_{a,235} \bar{N}_{235} \phi,$$

$$\frac{d\bar{N}_{238}}{dt} = -\sigma_{a,238} \bar{N}_{238} \phi,$$

$$\frac{d\bar{N}_j}{dt} = -\sigma_{a,j} \bar{N}_j \phi + \sigma_{c,j-1} \bar{N}_{j-1} \phi,$$

Using the relationship between an increment of fluence ($\phi \Delta t$) and an increment of burnup (ΔBu):

$$\frac{d\bar{N}_{235}}{dbu} = -\sigma_{a,235} \bar{N}_{235} A,$$

$$\frac{d\bar{N}_{238}}{dbu} = -\sigma_{a,238} \bar{N}_{238} A,$$

$$\frac{d\bar{N}_j}{dbu} = -\sigma_{a,j} \bar{N}_j A + \sigma_{c,j-1} \bar{N}_{j-1} A,$$

$$A = 0.8815 \frac{\rho_{fuel}}{\alpha \sum_k \sigma_{f,k} \bar{N}_k}$$

- 0.8815 = (238/270) - ratio of the mass of U in UO_2 to the mass of UO_2
- Burnup units - MW days per ton heavy metal
- σ_f is the fission cross section
- A - energy released per fission

NEUTRONICS MODELS IN FUEL PERFORMANCE CODES

TUBRNP model

Local isotope concentrations (as function of radial position):

$$\frac{dN_{235}}{dbu} = -\sigma_{a,235}N_{235}(r)A$$

$$\frac{dN_{238}}{dbu} = -\sigma_{a,238}\bar{N}_{238}\bar{f}(r)A$$

$$\frac{dN_{239}}{dbu} = -\sigma_{a,239}N_{239}(r)A + \sigma_{c,238}\bar{N}_{238}\bar{f}(r)A$$

$$\frac{dN_j}{dbu} = -\sigma_{a,j}N_j(r)A + \sigma_{c,j-1}N_{j-1}(r)A \quad - \quad j \text{ represents each of the } {}^{240}\text{Pu}, {}^{241}\text{Pu}, \text{ and } {}^{242}\text{Pu} \text{ isotopes}$$

$$\frac{2 \int_{r_{in}}^{r_{out}} \bar{f}(r)r dr}{r_{out}^2 - r_{in}^2} = 1 \quad \longrightarrow \quad \bar{f}(r) = \frac{f(r)}{\int_{r_{in}}^{r_{out}} f(r)r dr}$$

$$f(r) = 1 + p_1 e^{-p_2(r_{out}-r)^{p_3}}$$

NEUTRONICS MODELS IN FUEL PERFORMANCE CODES

TUBRNP model

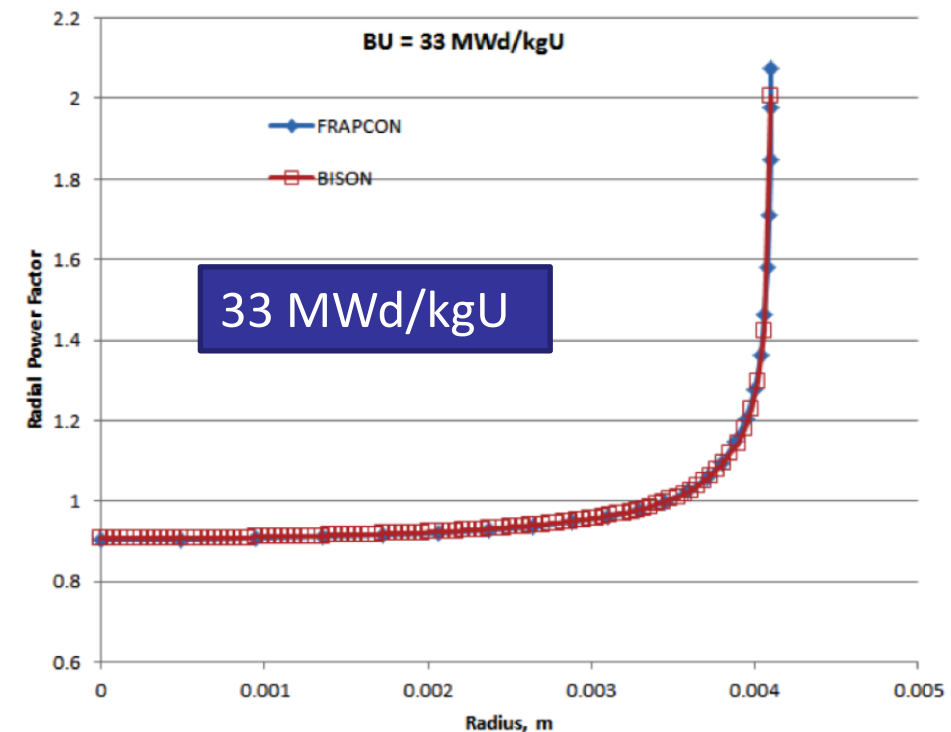
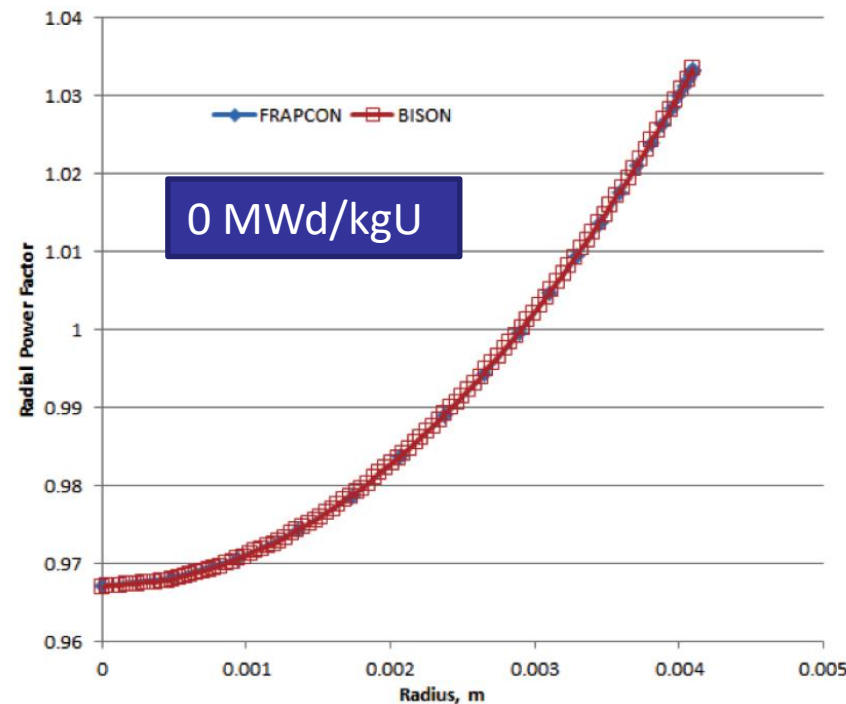
An approximate one-group 1D thermal diffusion equation with fixed cross-sections: $\nabla^2 \phi - k^2 \phi = 0$

For a solid cylinder, the neutron flux, $\phi(r)$, is a function of the modified Bessel functions:

$$\phi(r) = C_1 I_0(\kappa r)$$

$$\kappa = \sqrt{\frac{\Sigma_{a,tot}}{D}} \quad \Sigma_{a,tot} = \sum_k \sigma_{a,k} \bar{N}_k \quad D = \frac{1}{3\sigma_s \bar{N}_{tot}}$$

- I is the modified Bessel function
- C_1 is a constant
- κ is the inverse diffusion length



NEUTRONICS MODELS IN FUEL PERFORMANCE CODES

Burnup Calculation

- Burnup, which is used to calculate fuel properties and the fuel densification and swelling rates, is computed at each material or integration point as:

$$\beta = \frac{\dot{F} t}{N_f^0}$$

$$N_f^0 = \frac{\rho N_{av}}{M_w}$$

- $\dot{F}$ is the volumetric fission rate
- t is time
- N_f^0 is the initial density of heavy metal atoms in the fuel
- ρ is the initial fuel density
- N_{av} is Avagadro's number
- M_w is the molecular weight

- A burnup increment is computed for each time increment and summed to give the total burnup

$$\beta_i = \beta_{i-1} + \frac{\dot{F} \Delta t}{N_f^0}$$

Fission Rate

- The fission rate, (fission/m³/s), is calculated from the local power density:

$$\dot{F} = \frac{P}{\alpha}$$

- P is the power density (W/m³)
- α is the energy released per fission (J/fission)

Fast Neutron Flux

- Fast neutron flux may be specified as problem input; or estimated given the linear heat rate:

$$\dot{\Phi} = cP$$

- c is a conversion factor, typically $3e^{13}$ (n/(m²s)/(W/m))
- P is the linear heat rate (W/m)

Fast Neutron Fluence

- Fast neutron fluence is the time-integrated fast neutron flux:

$$\Phi_{n+1} = \Phi_n + \Delta t \dot{\Phi}$$

- Φ_n is the value of the fast neutron fluence at step n
- Δt is the timestep size

NEUTRONICS MODELS IN FUEL PERFORMANCE CODES

Decay Heat model - the “simplified method” of 1979 ANS-5.1 Standard on Decay Heat Power in LWR.

- !! The method assumes that the decay heat power from fissioning isotopes other than ^{235}U is identical to that of ^{235}U and that the fission rate is constant over the operating history at a maximum level corresponding to $P_{\max}$.
- This simplified method **overestimates decay heat power**, especially with respect to LWR cores having an appreciable amount of plutonium
- For a finite reactor operating time, the decay heat power is approximated as:

$$P_d(t, T) = P_{\max} \frac{1.02 G_n (F(t, \infty) - F(t + T, \infty))}{Q_{\text{mev}}}$$

- t is the time following reactor shutdown (s)
- T is the total operating time including intermediate periods at zero power (s)
- G_n is the neutron capture factor, Q_{mev} is the energy released per fission (MeV/fission)
- $F(t, \infty)$ is the decay heat power (MeV/fission) for thermal fission of ^{235}U for an infinite-time base irradiation

TH MODELS IN FUEL PERFORMANCE CODES

Example: BISON

- A single coolant channel is used mathematically to describe the thermal boundary condition for modeling the fuel rod behavior
- The model covers two theoretical aspects, i.e., **the local heat transfer from cladding wall into the coolant and the thermal energy deposition in the coolant in steady state and slow operating transient conditions**

Assumptions and limitations:

Closed channel - the lateral energy, mass, and momentum transfer in the coolant channel within a fuel assembly is neglected - axial direction only

Homogeneous and equilibrium flow - for two-phase flow, the thermal energy transport and relative motions between the two phases are neglected - one pseudo fluid

Fully developed flow - the heat transfer is assumed to happen in a condition that the boundary layer has grown to occupy the entire flow area, and the radial velocity and temperature profiles are well established

Pressure drop neglected - pre-calculated coolant pressure as a function of time and axial location is an input provided by user

TH MODELS IN FUEL PERFORMANCE CODES

Coolant Enthalpy Rise in a Single Channel

- In steady state operation, the enthalpy rise in a coolant channel with incompressible fluid can be derived using energy conservation equation:

$$H(z) = H_{in} + \frac{\int_0^Z q''(z) \pi D_h dz + \int_0^Z f_c q' dz}{GA}$$

- H_{in} is the coolant enthalpy at inlet in (J/kg) ;
- $H(z)$ is the coolant enthalpy at axial location z in (J/kg)
- z is axial location (m)
- q'' is fuel rod surface heat flux (W/m²)
- q' is fuel rod linear heat generation rate (W/m)
- f_c is the fraction of heat generated in the coolant by neutron and gamma rays (dimensionless)
- D_h is heated diameter (m)
- G is coolant mass flux (kg/sec-m²)
- A is flow area of the coolant channel (m²)

TH MODELS IN FUEL PERFORMANCE CODES

Coolant Enthalpy Rise in a Single Channel

$$H(z) = H_{in} + \frac{\int_0^Z q''(z) \pi D_h dz + \int_0^Z f_c q' dz}{GA}$$

- The mass flux, pressure, and coolant temperature at the inlet of coolant channel are provided as input for calculating coolant enthalpy rise
- With calculated enthalpy and input coolant pressure, the corresponding thermo-dynamic condition can be determined using a steam table
- The thermal-physical properties of water and steam are evaluated at the corresponding bulk coolant temperature or at the cladding wall temperature for the use of calculating heat transfer coefficients between the cladding wall and the coolant
- BISON input allows the mass flux, pressure, and coolant temperature to change as a function of time, although the coolant enthalpy is still solved using the same equation above
- Allowing the variation of inlet thermal-hydraulic conditions can be used to model a quasi-steady state when the velocity and thermal energy of coolant at a given location are assumed to achieve the equilibrium condition instantaneously

TH MODELS IN FUEL PERFORMANCE CODES

Pre-CHF Heat Transfer Correlations

Single phase liquid forced convection - Dittus-Boelter correlation

$$Nu = 0.023 Re^{0.8} Pr^{0.4}$$

Sub-cooled boiling - Jens-Lottes correlation

$$\Delta T = 25 \left(\frac{q''(z)}{10^6} \right)^{0.25} / e^{P/6.2 \times 10^6} \longrightarrow h = \frac{[(T_W - T_{sat}) e^{P/6.2 \times 10^6} / 25]^4 \times 10^6}{T_W - T_b}$$

Sub-cooled boiling - Thom correlation

$$\Delta T = 22.7 \left(\frac{q''(z)}{10^6} \right)^{0.5} / e^{P/8.7 \times 10^6} \longrightarrow h = \frac{[(T_W - T_{sat}) e^{P/8.7 \times 10^6} / 22.7]^2 \times 10^6}{T_W - T_b}$$

Saturated boiling - Shrock-Grossman correlation

$$h = \left(a_1 \frac{q''}{G h_{fg}} + a_2 X_{tt}^{-b} \right) h_l \quad X_{tt}^{-1} = \left(\frac{x}{1-x} \right)^{0.9} \left(\frac{\rho_f}{\rho_g} \right)^{0.5} \left(\frac{\mu_g}{\mu_f} \right)^{0.1}$$

Saturated boiling - Chen correlation

$$h_c = 0.023 \left(\frac{G(1-x)D_e}{\mu_f} \right)^{0.8} Pr_f^{0.4} \frac{k_f}{D_e} \quad F = 1, \text{ for } \frac{1}{X_{tt}} < 0.1$$

$$F = 2.35 \left(0.213 + \frac{1}{X_{tt}} \right)^{0.736}, \text{ for } \frac{1}{X_{tt}} > 0.1$$

$$h = F h_c + S h_{NB}$$

$$h_{NB} = 0.00122 \left[\frac{(k^{0.79} c_p^{0.45} \rho^{0.49})_f}{\sigma^{0.5} \mu_f^{0.29} h_{fg}^{0.25} \rho_g^{0.24}} \right] \Delta T_{sat}^{0.24} \Delta P^{0.75}$$

$$\Delta T_{sat} = T_W - T_{sat} \quad \Delta P = P(T_W) - P(T_{sat}) \quad S = \frac{1}{1 + 2.53 \times 10^{-6} Re^{1.17}}$$

TH MODELS IN FUEL PERFORMANCE CODES

CHF Correlations

PWRs - EPRI-Columbia correlation

$$\frac{q_{CHF}}{10^6} = \frac{A - x_{in}}{C + \left(\frac{x - x_{in}}{q''}\right)}$$

F_{AP} is the non-uniform axial heat flux distribution parameter:

$$F_{AP} = 1 + \frac{(Y - 1)}{(1 + G)}$$

Y is Bowring's non-uniform parameter defined as:

$$Y = \frac{\int_0^z q''(z) dz}{q''(z)z}$$

$$A = F_a p_1 p_r^{p_2} G^{(p_5 + p_7 p_r)}$$

$$C = F_c F_{AP} p_3 p_r^{p_4} G^{(p_6 + p_8 p_r)}$$

$$p_1 = 0.5328$$

$$p_2 = 0.1212$$

$$p_3 = 1.6151$$

$$p_4 = 1.4066$$

$$p_5 = -0.3040$$

$$p_6 = 0.4843$$

$$p_7 = -0.3285$$

$$p_8 = -2.0749$$

$$p_r = \text{critical pressure ratio} = \text{system pressure} / \text{critical pressure}$$

$$G = \text{local mass velocity (Mlbm/hr-ft}^2\text{)}$$

$$x_{in} = \text{inlet quality}$$

$$F_a = G^{0.1}$$

$$F_c = 1.183 G^{0.1}$$

$$F_a = F_c = 1 \text{ for no cold wall}$$

$$q'' = \text{local heat flux (MBtu/hr-ft}^2\text{)}$$

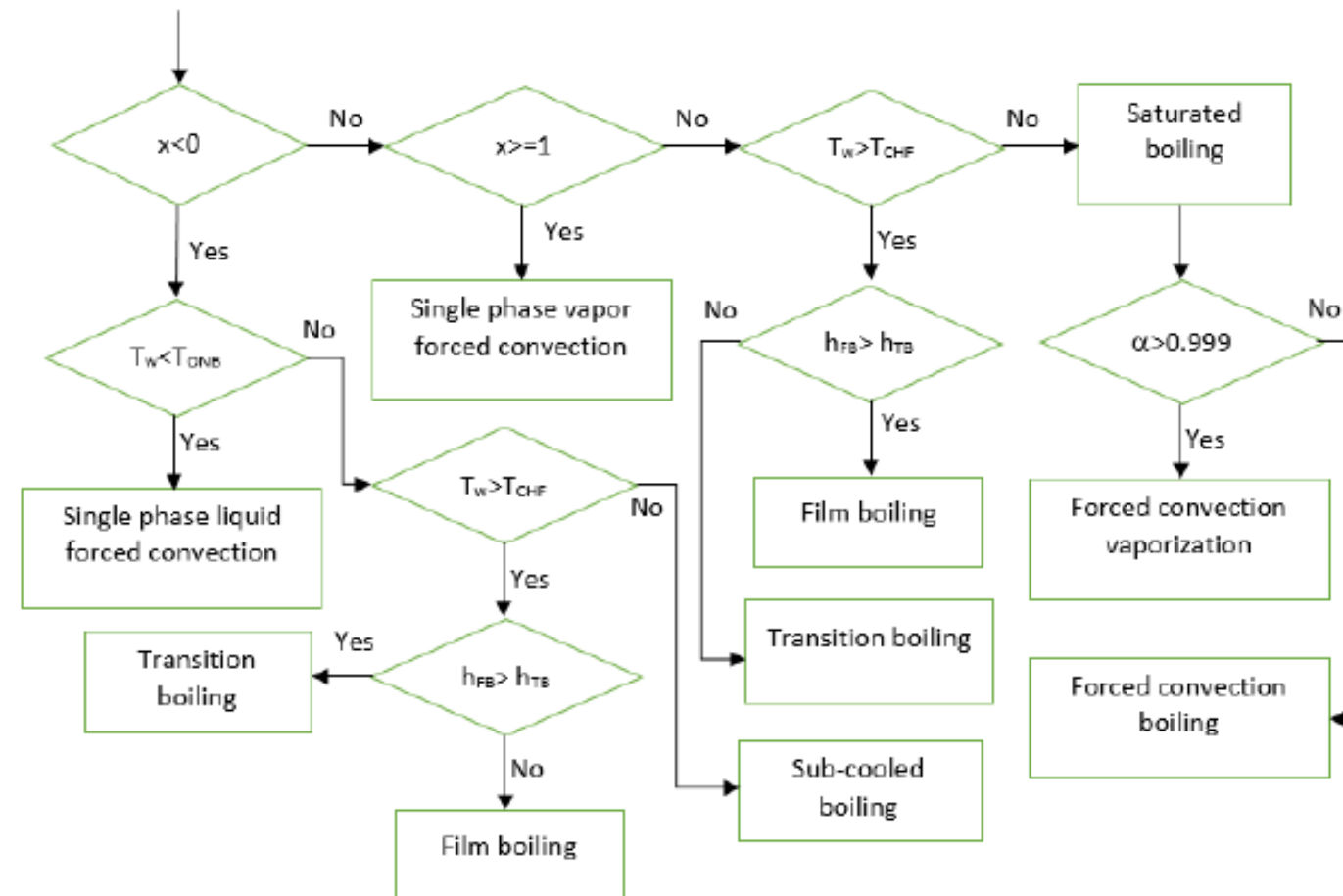
BWR - GE correlation

$$q_{CHF} = 10^6 (0.8 - x) \text{ for } G \geq 0.5 \times 10^6 \text{ lb}_m/\text{ft}^2\text{-hr}$$

$$q_{CHF} = 10^6 (0.84 - x) \text{ for } G < 0.5 \times 10^6 \text{ lb}_m/\text{ft}^2\text{-hr}$$

TH MODELS IN FUEL PERFORMANCE CODES

Heat transfer regimes selection criteria



T_{CHF} , temperature at critical heat flux
 T_{ONB} , temperature at the onset of nucleate boiling
 h_{FB} , heat transfer coefficient in the film boiling regime
 h_{TB} , heat transfer coefficient in the transition boiling regime
 x , steam quality
 α , void fraction

Water-Steam Properties - IAPWS-IF97 standard

HEAT CONDUCTION AND FUEL ROD MODELING IN SUBCHANNEL AND SYSTEM TH CODES

MODELING OF HEATED AND UNHEATED STRUCTURES

In system and sub-channel TH codes, the models for heated and unheated structures calculate the steady state and transient temperature response of the structure of interest, using the BCs and the fluid energy equation

This calculation may consist of five major components:

1. **HEAT GENERATION MODEL**, which determines the temporal and spatial variations in heat generation due to fission, gamma, and neutron energy deposition, fission product and actinide decay, and metal-water reaction in the cladding
2. **CONDUCTION MODEL**, which specifies the conductor geometry and material properties, and solves the conduction equation
3. **GAP CONDUCTANCE MODEL**, which is a dynamic gap conductance model for a nuclear fuel rod
4. **FUEL ROD DEFORMATION MODEL**, which calculates the deformation of nuclear fuel rods, and the effects on core thermal hydraulics
5. **QUENCH FRONT MODEL**, which is a “fine mesh-rezoning” method calculating the quench front propagation due to axial conduction and radial heat transfer

Thermal Phenomena Modeling

Calculation of the temperature distribution in the fuel rod

There are two general types of conductor models considered:

1. A model designed for structures containing a heat source such as [nuclear fuel rods](#) and [heater rods](#)

This model allows to simulate most of the conductor geometries found in a reactor vessel and heat transfer experiments

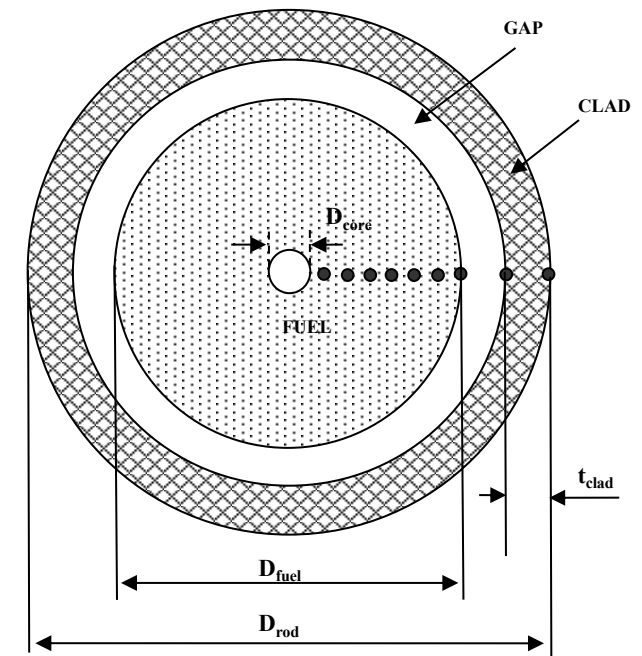
2. A model designed for [non-powered structures](#) such as [tubes](#) or [walls](#)

[Nuclear Fuel Rod](#)

Fuel and clad material properties are usually specified by input or obtained from tables of fuel & cladding properties

The properties are calculated using correlations (for example, from MATPRO-11)

Solid rods (or different geometry) with fluid thermal connections on the rod exterior are considered by this model



Thermal Phenomena Modeling

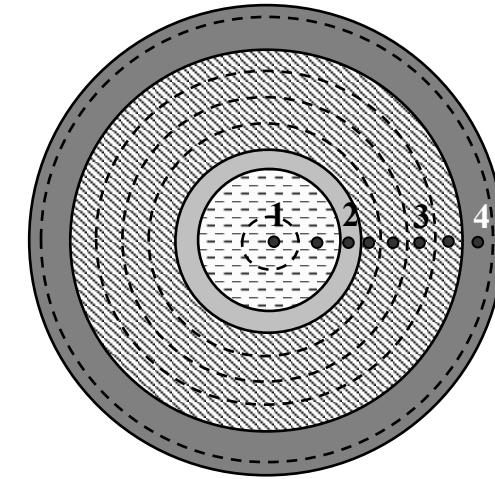
Calculation of the temperature distribution in the fuel rod

Electric Heater Rods

These rods consist of concentric rings of different material regions

In each region the material type, number of radial nodes, width, and power factor are specified by input

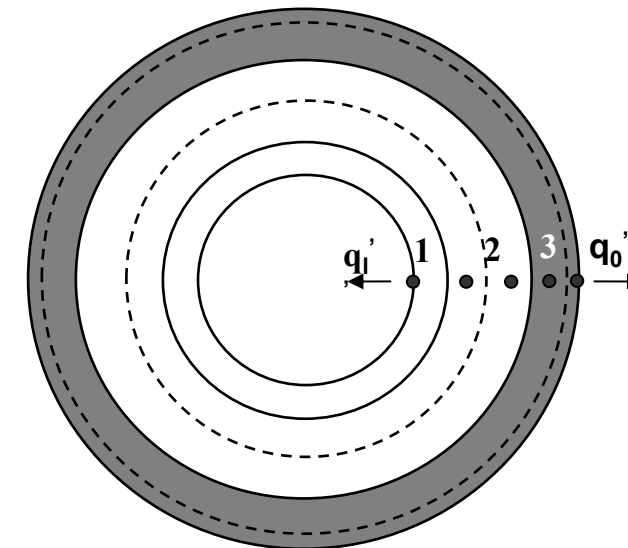
Contact resistances can be modeled by including a region one node wide with the material properties that give it the appropriate thermal resistance



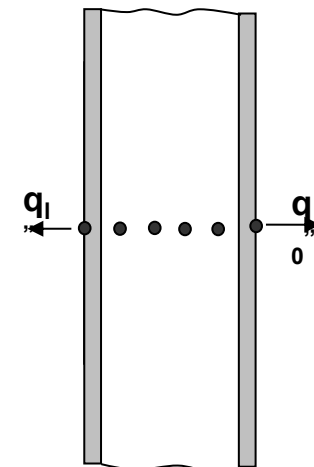
Tubes and Plates

Conductors, either tubes or plates, with thermal connections to channels on either the inner or outer surface are modeled by the tube and wall models

Concentric and flat plate fuel elements, thermal walls, and simple tubes can be modeled with these options



a) Tube



b) Wall

Thermal Phenomena Modeling

Calculation of the temperature distribution in the fuel rod

- Heat conduction equation is solved in the fuel with heat source coming from fission (power) and decay
- Heat conduction equation is solved in the cladding with potential heat source from metal-water reaction
- Pellet-cladding heat transfer modelling to calculate the temperature difference in the gap
- Cladding to coolant heat transfer to model the cladding external temperature

$$\int_V \rho(\vec{r}, T) c_p(\vec{r}, T) \frac{dT}{dt} dV = \int_s k(\vec{r}, T) \vec{\nabla} T \cdot d\vec{s} + \int_V q(\vec{r}) dV$$

$$\left. \frac{dT}{dr} \right|_{r=0} = 0, \quad \left. \frac{dT}{dr} \right|_{r=r_o} = T_s$$

where

ρ = density [kg/m³]

c_p = specific heat capacity [J/kg/K]

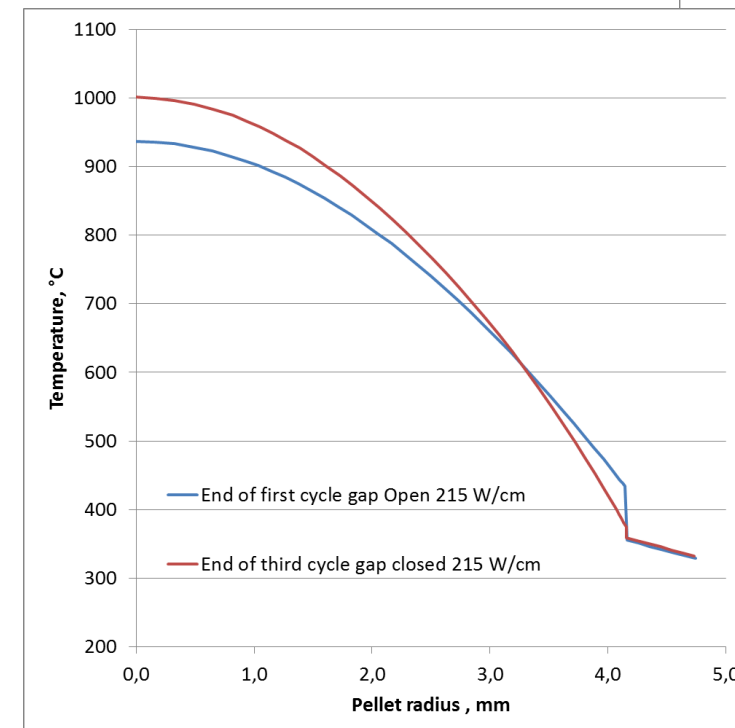
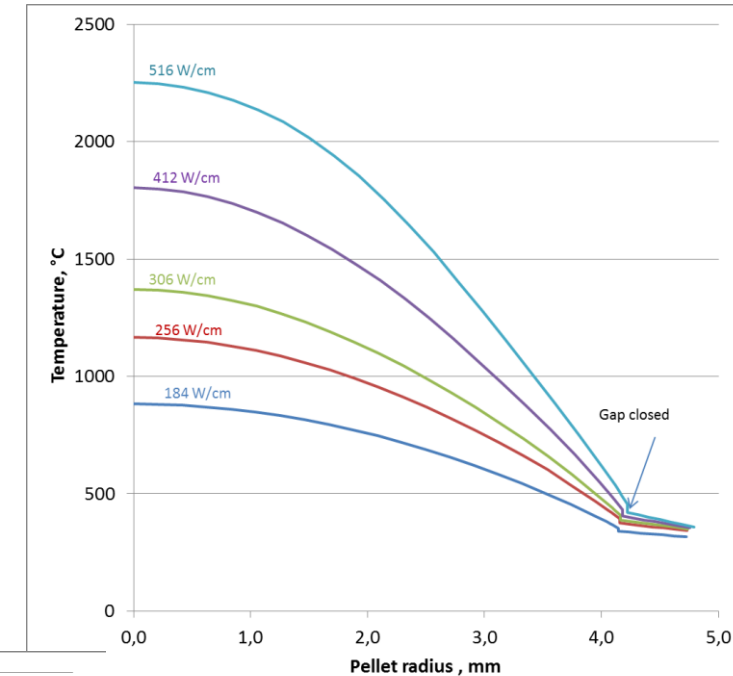
T = temperature [K]

V = control volume [m³]

k = thermal conductivity [W/m/K]

s = control volume surface [m²]

q = power density [W/m³]



Thermal Phenomena Modeling

Calculation of the temperature distribution in the fuel rod

Finite difference solution of fuel heat transfer equation

Radial power profile based on simplified neutronics depletion

Gap conductance

$$h = h_r + h_{gas} + h_{solid}$$

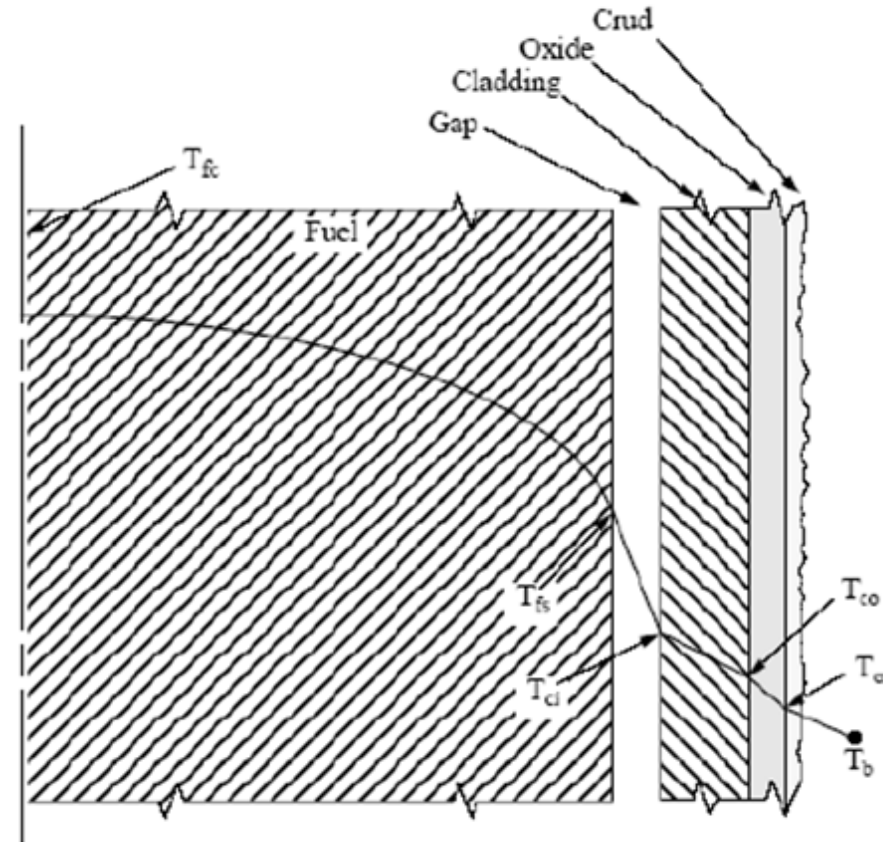
$$h_r = \sigma F [T_{fs}^2 + T_{cl}^2] [T_{fs} + T_{cl}]$$

$$h_{gas} = \frac{\nu_{gas}}{\Delta x}$$

$$h_{solid} = \frac{0.4166 K_m P_{rel} R_{mult}}{RE}, \text{ if } P_{rel} > 0.003$$

$$h_{solid} = \frac{0.00125 K_m}{RE}, \text{ if } 0.003 > P_{rel} > 9 \times 10^{-6}$$

$$h_{solid} = \frac{0.4166 K_m P_{rel}^{0.5}}{RE}, \text{ if } P_{rel} < 9 \times 10^{-6}$$



Dittus-Boelter film conductance

$$h_f = \left(\frac{0.023 k}{D_e} \right) \text{Re}^{0.8} \text{Pr}^{0.4}$$

Bulk coolant temperature

$$\iint_S k(T, \vec{x}) \vec{\nabla} T(\vec{x}) \cdot \vec{n} ds = \iiint_V S(\vec{x}) dV \leftarrow \Delta T_{gap} = \frac{q''(z)}{h} \leftarrow \Delta T_c = q''(z) r_o \ln(r_o / r_i) / k_c \leftarrow \Delta T_{ox}(z) = \frac{q''(z) \delta_{ox}(z)}{k_{ox}} \leftarrow \Delta T_{cr}(z) = q''(z) \frac{\delta_{cr}}{k_{cr}} \leftarrow \Delta T_f(z) = q''(z) / h_f \leftarrow T_b(z) = T_m + \int_0^z \left[\frac{(\pi D_o) q''(z)}{C_p G A_f} \right] dz$$

Solution of cladding heat transfer equation

Empirical oxide thickness and conductivity

Empirical crud thickness and constant conductivity

$$K_{gs} = \frac{1}{A + a \cdot gad + BT + f(Bu) + (1 - 0.9 \exp(-0.04Bu)) g(Bu) h(T) + \frac{E}{T^2} \exp\left(-\frac{F}{T}\right)}$$

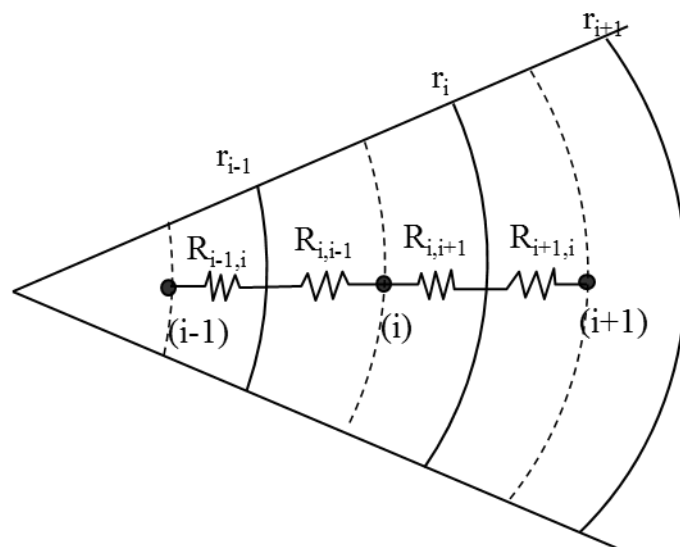
Thermal Phenomena Modeling

Calculation of the temperature distribution in the fuel rod

To simulate generalized conductor geometries and temperature-dependent material properties, one option is to employ [a finite difference form of the conduction equation](#):

- The finite-difference nodes of the conduction equation are modeled as control volumes connected by thermal resistances
- They form a set of linearized equations, which are solved by Gaussian elimination and back-substitution
- The radial conduction equation for a control volume are derived from a simple heat balance

For node i



$$\begin{aligned}
 (\rho C_p V)_i \frac{\partial T_i}{\partial t} = & - \underbrace{Q_{i,i-1}}_{\text{radial heat flow from node (i) to (i-1)}} - \underbrace{Q_{i,i+1}}_{\text{radial heat flow from node (i) to (i+1)}} + \underbrace{Q_i''' V_i}_{\text{volumetric heat generation rate}}
 \end{aligned}$$

Thermal Phenomena Modeling

Calculation of the temperature distribution in the fuel rod

- Heat transfer through a node is computed from the conductance K of the material and the temperature gradient across the node as:

$$Q_{j,i} = K_{j,i} (T_j - T_i) \quad \text{for } j = \begin{cases} i-1 \\ i+1 \end{cases}$$

and

$$K_{i,j} = K_{j,i} \quad \text{for } j = \begin{cases} i-1 \\ i+1 \end{cases}$$

- The conductance is the inverse of the thermal resistance, R , between nodes and is:

$$K_{i,i-1} = \frac{1}{R_{i,i-1} + R_{i-1,i}}$$

- Thermal resistances are calculated for each node as a function of geometry and thermal conductivity

Thermal Phenomena Modeling

Calculation of the temperature distribution in the fuel rod

- Once the thermal resistances have been characterized from the appropriate geometry formula and material properties, the conduction equation can be written as:

$$(MC_p)_i \frac{\partial T_i}{\partial t} = K_{i,i-1} (T_{i-1} - T_i) + K_{i,i+1} (T_{i+1} - T_i) + Q_i''' V_i$$

- Forward differencing the temporal derivatives, the above equation becomes:

$$\frac{(MC_p)_i}{\Delta t} (T_i - T_i^n) = K_{i,i-1} (T_{i-1} - T_i) + K_{i,i+1} (T_{i+1} - T_i) + Q_i''' V_i$$

where T_i^n is the temperature at the last time step

This equation applies to all internal nodes and variations of it are defined for the boundary nodes

The boundary conditions applied to the conduction equation can be adiabatic or a surface heat transfer coefficient:

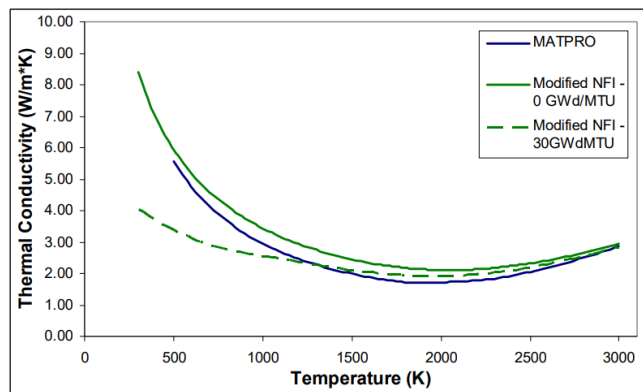
- 1. Adiabatic boundary conditions are assigned to the center nodes of solid rods (nuclear and cylindrical rods) and any surface node of tubes or walls not connected to the fluid*
- 2. Heat transfer coefficient boundary conditions are applied at any surface connected to the fluid*

Thermal Phenomena Modeling

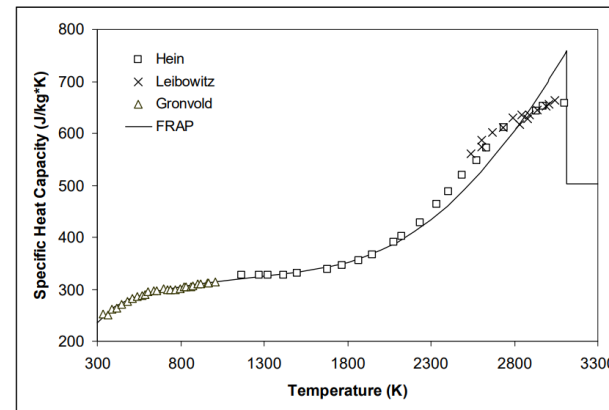
Calculation of the temperature distribution in the fuel rod

- Solution of steady state and transient heat conduction equation in the fuel and cladding
- Various correlations for thermal conductivities and specific heat capacities
- Usually, crud and oxide layers are modelled through specific thermal conductivities

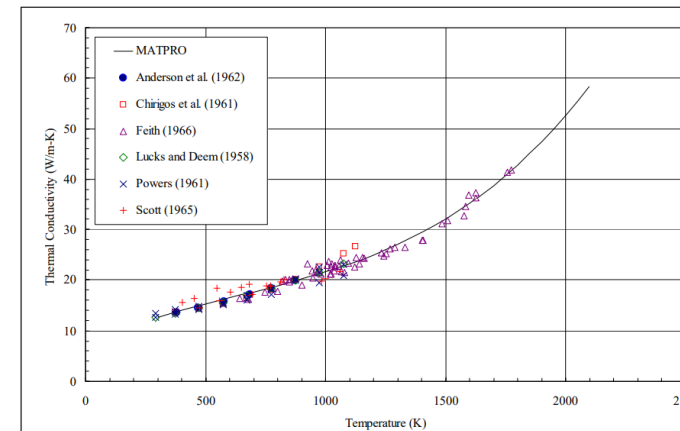
fuel thermal conductivity



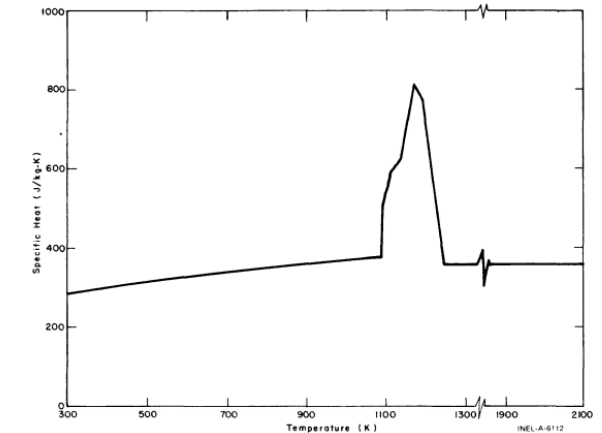
fuel specific heat capacity



cladding thermal conductivity



cladding specific heat capacity

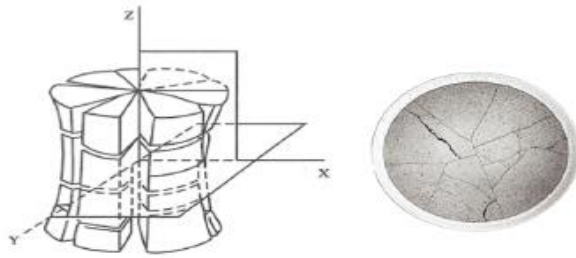


Mechanical Phenomena Modeling

Calculation of fuel rod deformations

Fuel deformation

- **Thermal Expansion**
- **Cracking and Relocation:** Directly from the first ramp to full power, the large temperature gradient creates cracks in the fuel decreasing the effective gap



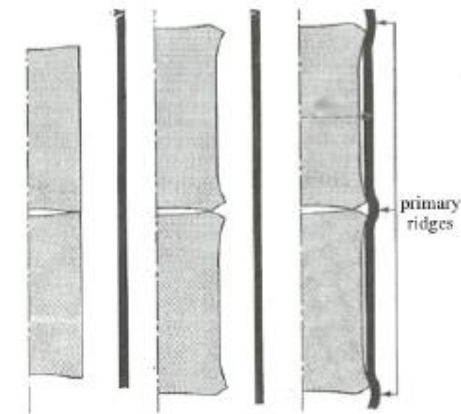
- **Densification:** The fuel due to fabrication process has initial porosity that decreases reducing the effective gap. The process stops around 5GWd/t
- **Creep**
- **Swelling:** Solid and gaseous fission products dislocate atoms and accumulate in the grain boundaries increasing the fuel volume and decreasing the effective gap
- **Stress induced elastic and plastic strains**

Cladding deformation

- **Thermal Expansion**
- **Creep:** Slow time dependent irreversible deformation process under the influence of stresses below the yield stress and has various origins, mechanical, thermal or irradiation. The most important impact is from the differential pressure leading to decrease of the gap
- **Stress induced elastic and plastic strains**

Pellet Cladding Mechanical Interaction (PCMI)

The gap immediately decreases due to relocation, then it increases due to densification and then progressively decreases due to fuel thermal expansion, swelling and cladding creep. The gap closes around 15GWd/t



Mechanical Phenomena Modeling

Calculation of fuel rod deformations

Assumptions

- Rigid pellet: no stress-induced deformation in fuel pellet.
- Small deformations.
- When gap is closed the fuel deformation is imposed on the cladding as well.

Cladding deformation

Irradiation and thermal creep are functions of stress, temperature, time and fast neutron flux

$$\varepsilon^c = f(\sigma, T, t, \phi)$$

Fuel deformation

Thermal expansion

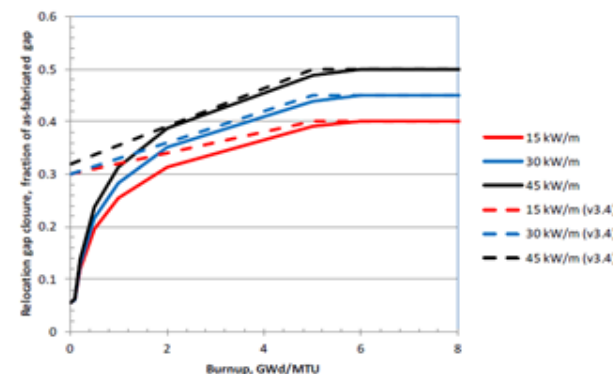
$$R_H = \sum_{i=1}^N \Delta r_i [1 + \alpha_{T_i} (T_i - T_{ref}) + \varepsilon_i^s + \varepsilon_i^d]$$

Gaseous swelling

$$\frac{\Delta l}{l} = 4.55 \times 10^{-5} T - 4.37 \times 10^{-2} \quad (960^\circ < T < 1370^\circ \text{C})$$

$$\frac{\Delta l}{l} = -4.05 \times 10^{-5} T + 7.40 \times 10^{-2} \quad (1370^\circ < T < 1832^\circ \text{C})$$

Densification and relocation are functions of Bu with full densification at around 5GWd/t. Relocation is shown in the Figure:



Plastic deformation

$$\sigma_\theta = \frac{r_i P_i - r_o P_o}{t}$$

$$\sigma_z = \frac{r_i^2 P_i - r_o^2 P_o}{r_o^2 - r_i^2} \quad \text{II}$$

$$\varepsilon_{ij} = \frac{1+\nu}{E} \sigma_{ij} - \delta_{ij} \left(\frac{\nu}{E} \sigma_{kk} - \int \alpha dT \right) + \varepsilon_{ij}^p + d\varepsilon_{ij}^p$$

Compatibility

$$\varepsilon_{ij,kl} + \varepsilon_{kl,ij} - \varepsilon_{ik,jl} - \varepsilon_{jl,ik} = 0$$

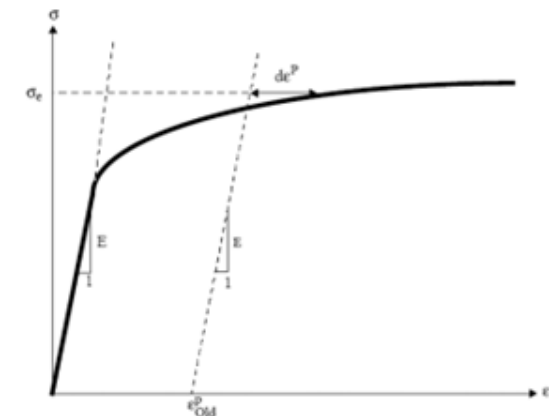
Definitions used in plasticity

$$\sigma_e = \sqrt{\frac{3}{2} S_{ij} S_{ij}}$$

$$S_{ij} = \sigma_{ij} - \frac{1}{3} \sigma_{kk} \delta_{ij}$$

Prandtl-Reuss flow rule

$$d\varepsilon_{ij}^p = \frac{3}{2} \frac{d\varepsilon^p}{\sigma_e} S_{ij}$$

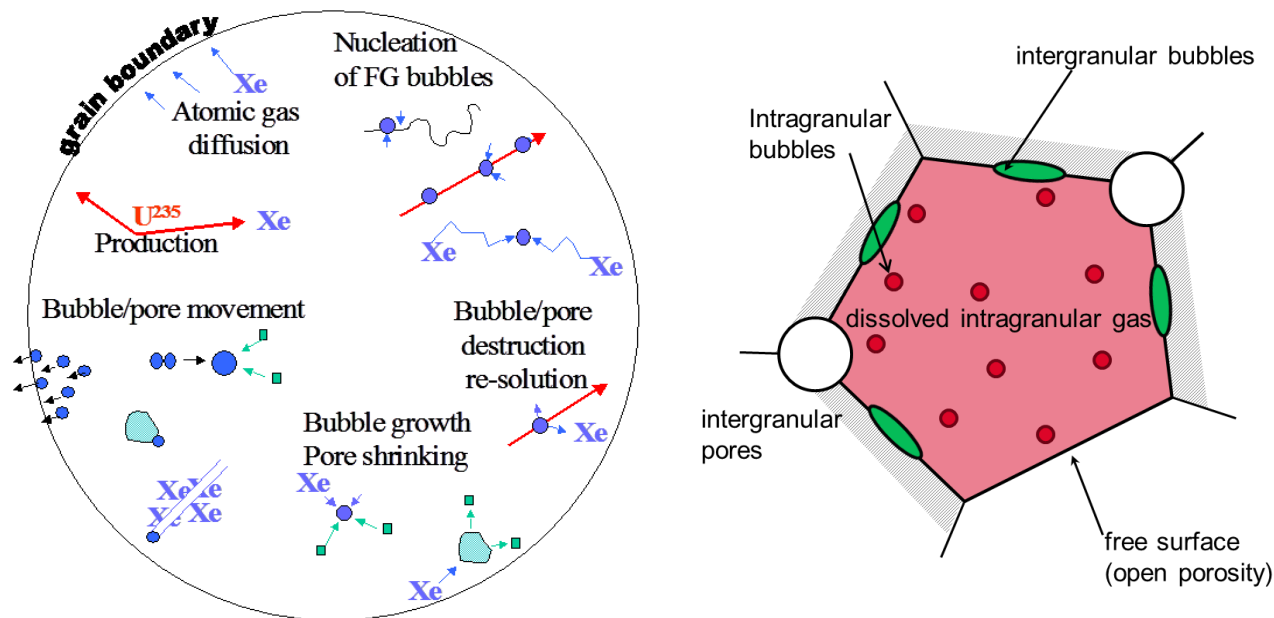


Physico-chemical Phenomena Modeling

Calculation of gas pressure and cladding oxidation

Fission Gas Production and Release

- Both solid and gas fission products are produced from the BOL
- The fission products distort the atom structure knocking atoms and creating vacancies. This reduces the fuel conductivity
- Fission gas products form bubbles that diffuse to grain boundaries, create tunnels and eventually are released in the gap, increasing the gap pressure and decreasing the gap conductance

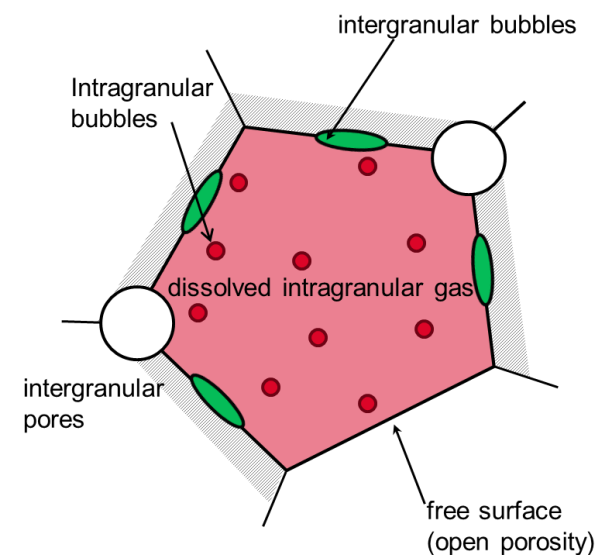


Radial Power and High Burnup Structure (HBS)

- Epithermal neutrons are captured close to the periphery by U^{238} increasing the local Pu^{239} concentration. This increases the local power generation and local burnup
- The high local burnup in combination with the low temperature at the periphery leads to the formation of the high burnup structure (HBS). This zone is formed around 60GWd/t (local), has very large porosity, very small grains ($10\mu m \rightarrow 0.2\mu m$) and contains many fission gas products that could be released in the gap

Cladding Waterside Corrosion

- Cladding oxidation leads to the formation of an outer zirconium oxide with low conductivity
- This reaction releases hydrogen in the cladding that can precipitate and render the cladding more brittle and prone to failure in transients



Physico-chemical Phenomena Modeling

Calculation of gas pressure and cladding oxidation

Oxidation thickness

The thickness is computed based on the oxidation rate.

The parameters of the equation depend on the cladding material.

$$\frac{d(s^3)}{dt} = \frac{A}{s^2} \exp\left\{\frac{-Q_1}{RT_1}\right\}$$

Hydrogen pickup

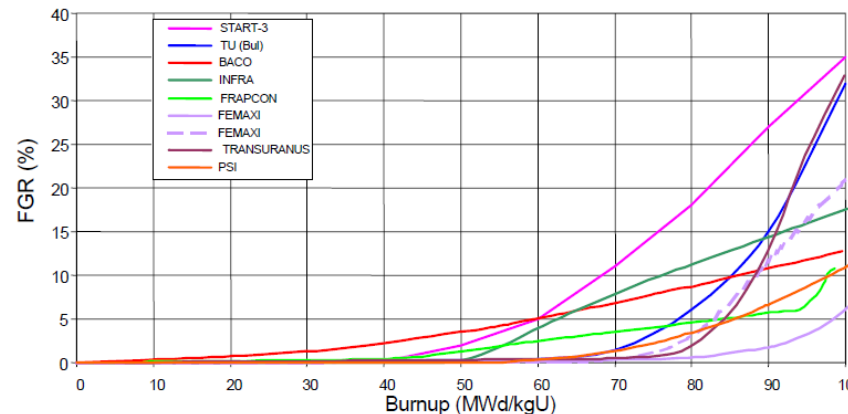
In PWR, a constant pickup fraction is assumed depending on the cladding material. In BWR a function of burnup is considered.

Gas Pressure

$$P = \frac{MR}{\frac{V_p}{T_p} + \sum_{n=1}^N \left[\frac{V_g}{T_g} + \frac{V_{ch}}{T_{ch}} + \frac{V_{cr}}{T_{cr}} + \frac{V_{dsh}}{T_{dsh}} + \frac{V_{por}}{T_{por}} + \frac{V_{rf}}{T_{rf}} + \frac{V_i}{T_i} \right]}$$

Volumes calculation

- Dish and Chamfer volumes.
- Gap volume.
- Fuel crack volume.
- Plenum volume.
- Open porosity volume.
- Roughness volume.



Fission Gas production

$$GPT(z) = \frac{BU(z)VF(z)}{100A_v} (PR_{krypton} + PR_{helium} + PR_{xenon})$$

Modified Forsberg-Massih Fission Gas release

Diffusion of produced fission gas to grain boundaries

$$\frac{dC}{dt} = D(t)\Delta_r C(r,t) + \beta(t)$$

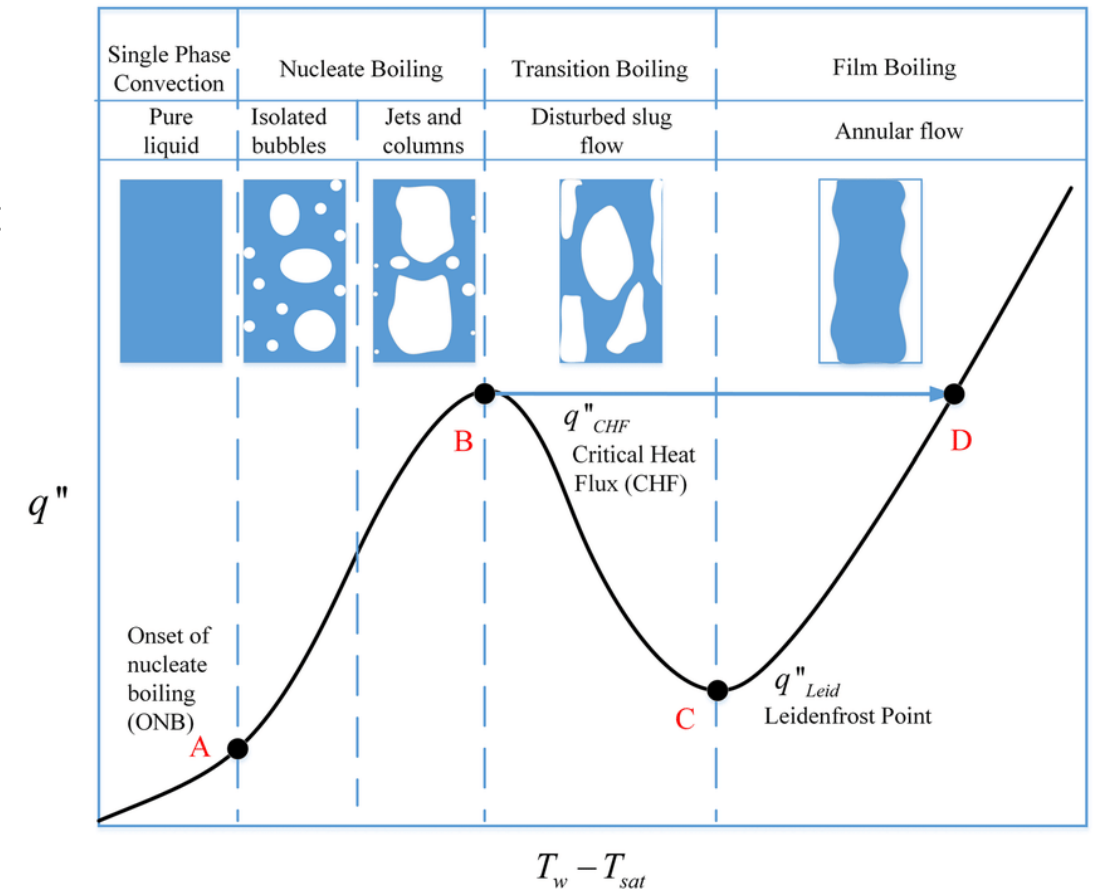
The gas is accumulated at the boundaries until a saturation concentration is achieved, then it is released

$$N_s = \left[\frac{4rF(\theta)V_c}{3K_B T \sin^2(\theta)} \right] \left(\frac{2\gamma}{r} + P_{ext} \right)$$

Fuel Rod Quench Front Modeling

The fuel rod models have several features to handle the unique situation that arise with heat conductors

- Thermal-hydraulic numerical simulations of rewetting encounter difficulties with large axial computational mesh spacing which cannot adequately resolve the axial profile of temperature and surface heat flux across the quench front
- During quenching, the entire boiling curve - from film boiling through transition boiling and critical heat flux to nucleate boiling - can be encompassed by one hydrodynamic mesh cell
- Constraining the entire cell to be in one boiling regime is non-physical and results in stepwise cell-by-cell quenching, producing flow oscillations that can obscure the correct hydrodynamic solution
- Consequently, an integration of the boiling curve shape through the hydrodynamic computational cell must be performed to determine the fluid heat input



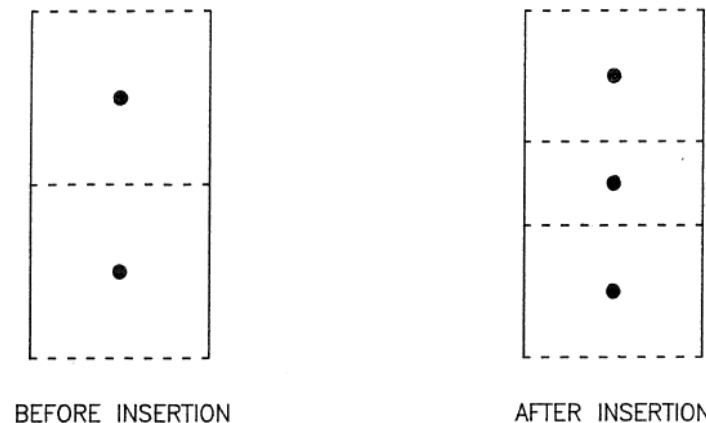
Fuel Rod Quench Front Modeling

User-specified fixed transfer nodes and a fine mesh-rezoning technique are employed to surmount these difficulties:

- Fixed heat transfer nodes are used to model fuel rod conduction and cladding-fluid heat transfer with nodes that are smaller than the hydrodynamic cell size
- These nodes are used throughout the transient - the fine mesh rezoning option allows the code to further resolve the quench front heat transfer
- Fine mesh heat transfer cells with axial and radial conduction are superimposed upon the fixed heat transfer nodes, and boiling heat transfer package is applied to each node
- By solving the two-dimensional conduction equation for a variable fine mesh at the quench front, propagation due either to quenching or dryout can be resolved and the surface heat flux integrated to provide the cell-averaged phasic heat inputs for the fluid energy equation
- The resulting quench front velocity will be a function of :
 1. *axial conduction*
 2. *boiling curve shape*
 3. *pre-quench heat transfer*
 4. *internal heat transfer within the rod*

Fuel Rod Quench Front Modeling

- Resolution of axial temperature and surface heat flux excursions is achieved by rezoning the heat conductor mesh in their vicinity
- When axial temperature differences between adjacent heat transfer nodes within a continuity cell exceed splitting criteria, an additional node is inserted halfway between the two original nodes
- When the splitting criteria are exceeded at continuity cell boundary, one node is inserted below the boundary and another is inserted above the boundary:



- Conversely, when a fine mesh has been established, but the disturbance had propagated downstream and the fine mesh is no longer necessary, adjacent nodes can be coalesced back down to one node

Dynamic Gap Conductance Model

In the dynamic gap conductance models, the gap thermal and geometric properties are calculated taking into account the thermal expansion of the fuel pellet and cladding, and the elastic deformation of the cladding

- The dimensions of the gap and the thermodynamic properties of the fill gas are used to calculate the gap conductance
- The fuel relocation model allows fuel to move radially into the fuel-cladding gap
- Cracks formed in the fuel by relocation reduce the effective conductivity of the fuel; both radial relocation and conductivity degradation are included in the calculation

The pellet-cladding gap conductance has three components:

$$h_{gap} = h_{rad} + h_{gas} + h_{solid}$$

h_{rad} = gap conductance due to thermal radiation

h_{gas} = gap conductance due to conduction in fill gas

h_{solid} = gap conductance due to physical conductance between fuel pellet and cladding

Dynamic Gap Conductance Model

Radiant Heat Transfer

The gap conductance due to radiant heat transfer is the ratio of the gap radiant heat flux to the temperature rise across the fuel/cladding gap:

$$h_{rad} = \frac{q_r''}{T_1 - T_2}$$

The radiant heat flux level leaving the fuel surface is determined from the Stefan-Boltzman equation using appropriate fuel cladding geometry factors, so that:

$$q_r'' = \sigma_{SB} \left[\frac{1}{\varepsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\varepsilon_2} - 1 \right) \right]^{-1} [T_1^4 - T_2^4]$$

where

- A_1 is the fuel surface area (ft²)
- A_2 is the cladding surface area (ft²)
- ε_1 is the fuel surface emissivity
- ε_2 is the cladding surface emissivity
- T_1 is the fuel surface temperature (°R)
- T_2 is the cladding surface temperature (°R)
- σ_{SB} is the Stefan-Boltzman constant (1.714×10^{-9} Btu/hr-ft²-°R⁴)

Dynamic Gap Conductance Model

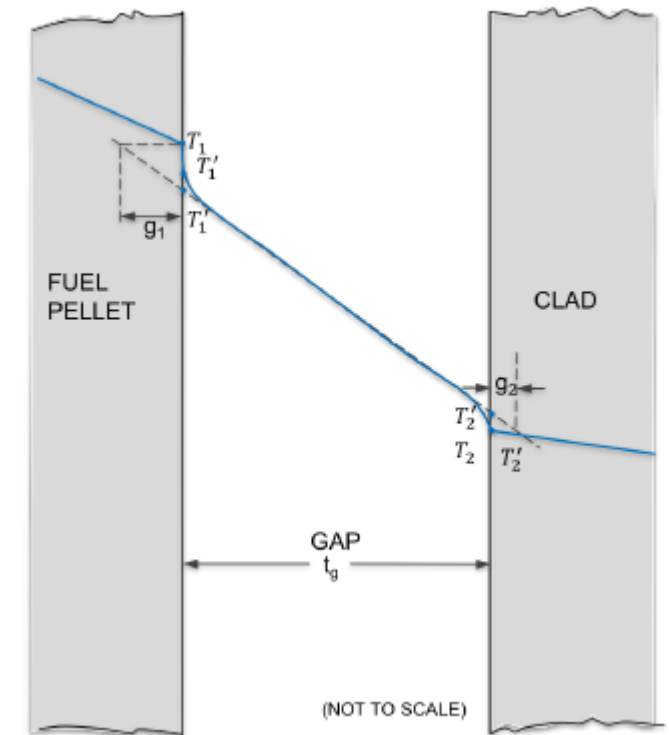
Conduction Heat Transfer in the Fill Gas

Heat conduction through the fill gas is often calculated using the Ross-Stoudt model:

For a normal open gap, the conductance:

$$h_{gas} = \frac{k_{gas}}{\tau_g + 1.8(g_1 + g_2)}$$

where k_{gap} is the fill gas mixture thermal conductivity (Btu/hr-ft-°F)
 τ_g is the gas gap width from deformation model (ft)
 g_1 is the fuel pellet temperature jump distance (ft)
 g_2 is the cladding temperature jump distance (ft)



- The temperature jump distances compensate for the nonlinearity of the temperature gradient near the walls and the temperature discontinuities on the wall surface
- The nonlinear temperature gradient is due to the incomplete thermal mixing of the gas molecules near the surface
- The surface temperature discontinuity results from the incomplete thermal accommodation of the gas molecule to the surface temperature

Dynamic Gap Conductance Modeling

Conduction Heat Transfer in the Fill Gas

When fuel/cladding contact occurs, the heat conductance in the gas becomes:

$$h_{gas} = \frac{k_{gas}}{\{1.8[C(R_1 + R_2) + g_1 + g_2] - 4.2(10^{-7})\}}$$

with $C = 1.98 \exp [-8.8(10^{-5}) P_i]$; R_1 and R_2 are the mean surface roughness of fuel pellet and the mean inside surface roughness

Fuel cladding contact is defined to occur when: $\tau_g \leq 3.6 (R_1 + R_2)$

Dynamic Gap Conductance Modeling

Pellet-Cladding Contact Conductance

When the fuel and cladding are not in contact, h_{solid} must be zero

But when the deformation model determines that the gap between the fuel and cladding is small enough for contact to occur, the Mikic/Todreas model is used to determine the contact conductance:

$$h_{\text{solid}} = \frac{5k_m}{(R_1^2 + R_2^2)^{1/2}} \left(\frac{P_{\text{int}}}{H_M} \right)^n \left(\frac{R_1}{\lambda_1} \right)$$

where $k_m = \frac{2k_1k_2}{k_1 + k_2}$

k_1 is the fuel thermal conductivity (Btu/hr-ft-°F)

k_2 is the cladding thermal conductivity (Btu/hr-ft-°F)

R_1 is the mean fuel surface roughness (in)

R_2 mean cladding surface roughness (in)

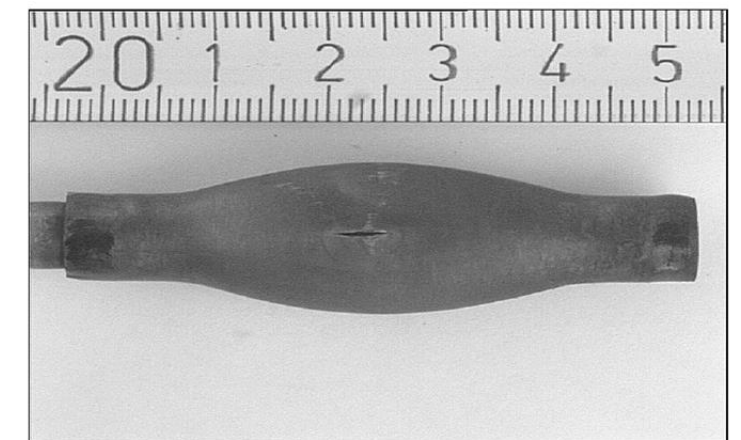
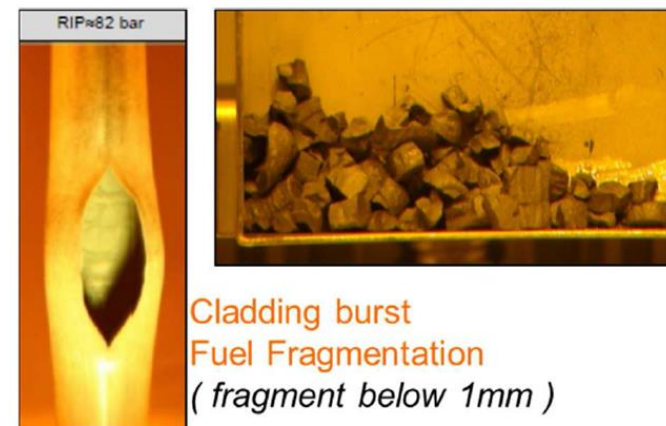
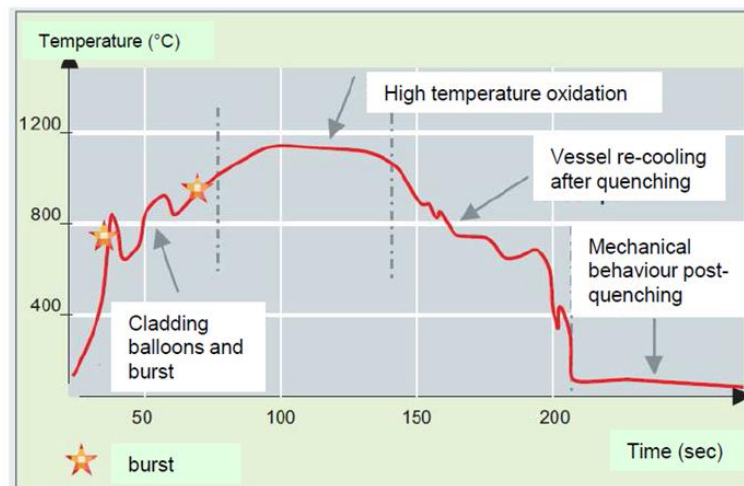
Accidental Conditions and Safety

■ LOCA

- Cladding temperature increase
 - Ballooning and burst
- At high burnup, fuel fragmentation and relocation (FFRD) due to over-pressurized FG bubbles

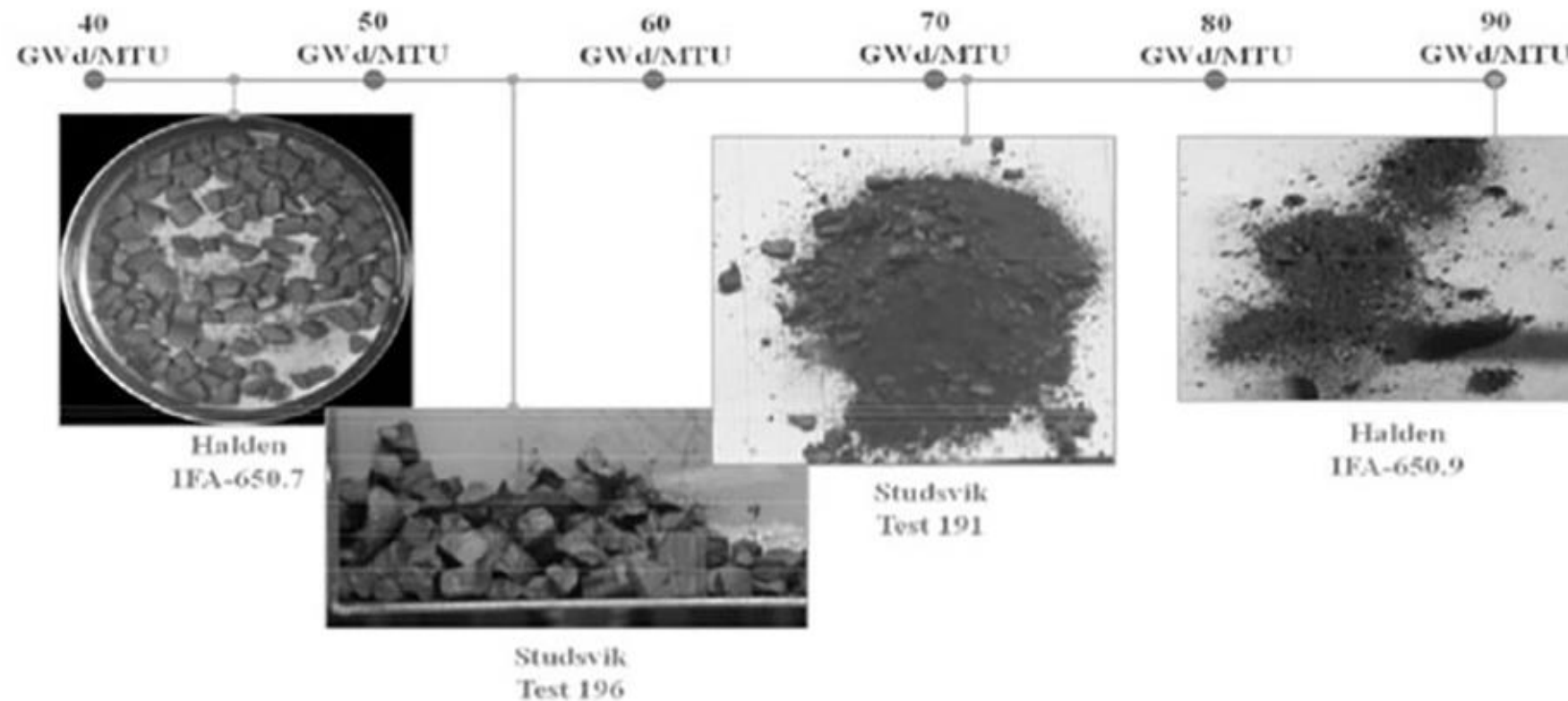
■ RIA

- *Pellet-cladding mechanical interaction (PCMI)*
 - Gas bubbles swelling → pellet volume increase
- *Clad ballooning*
 - FG release increase
 - Higher cladding temperature → plasticity



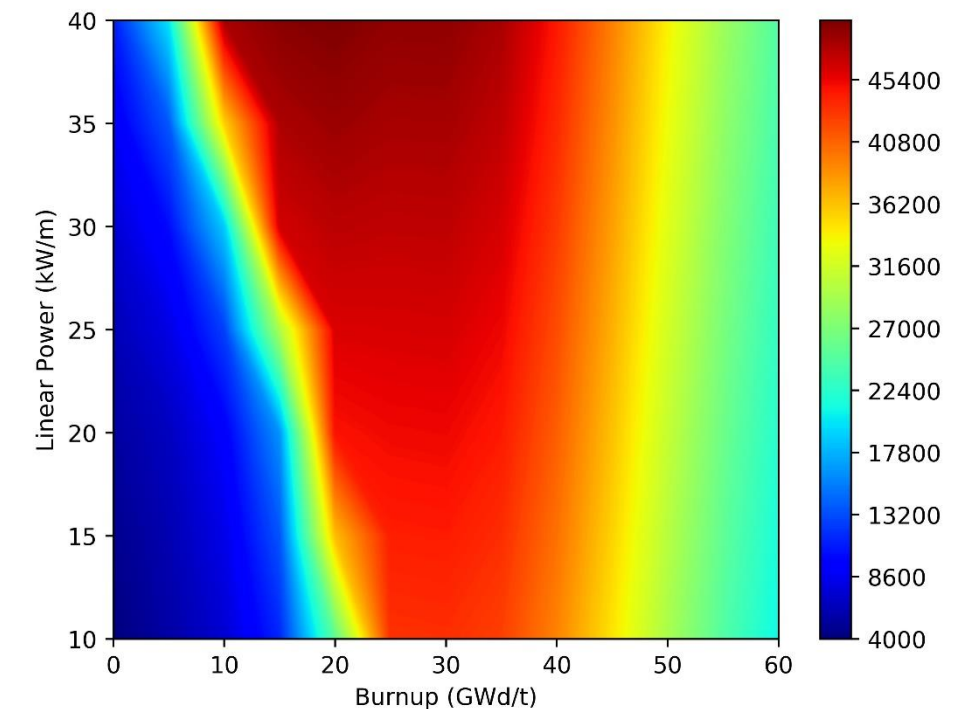
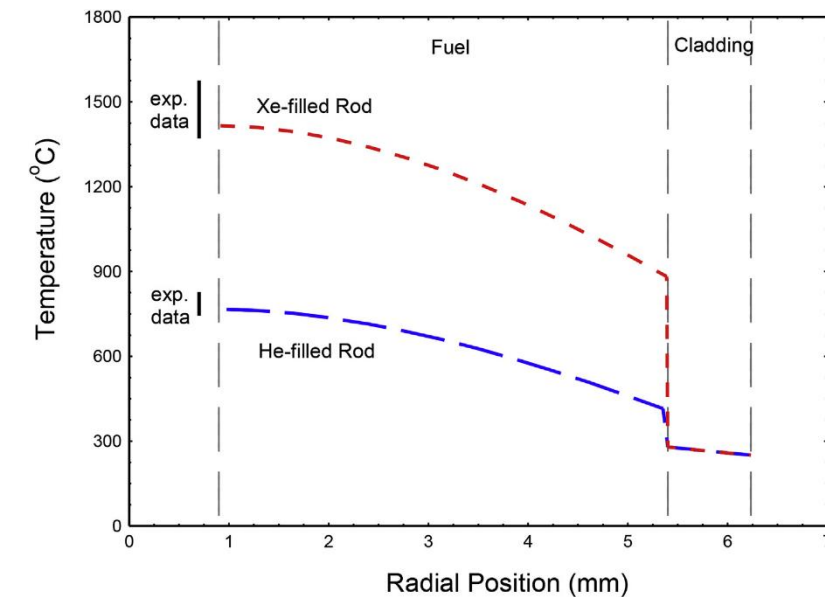
Accidental Conditions and Safety

- Fuel melting due to high temperatures
- Cladding melting
- Low temperature cladding burst
- High temperature cladding ballooning and burst
- Fuel Fragmentation, Relocation, and Dispersal (FFRD) for high burnup fuel



Questions ?

- What is the effect of the fission gas release on fuel temperature ?
- If fuel rod irradiation at constant power of 20kW/m reaches 60GWd/t in 2 years, how much time will it take for a fuel rod irradiation at 10kW/m constant power to reach the same burnup ?
- In which of the previous cases the gap will close sooner and why ?
- If the value of the gap heat transfer is H1 at 5GWd/t, how the value H2 at 30GWd/t compares to H1? How the value H3 at 60GWd/t compares to H2 ?



Next Class:

Space-Energy Dependent Dynamics: General Discussion of the Dynamics Problem; Flux Factorization